Lecture 2 Homework

Programming

None

Exercises

1. Prove that in any group for any elements $h, x, y \in G$ we have $h(xy)h^{-1} = (hxh^{-1})(hyh^{-1})$.
2. Make an addition table for Z_4.
3. Why isn't $\{\bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$ a group under addition or multiplication, mod 6?
4. Prove by induction that in an Abelian group, G we have $(gh)^n = g^n h^n$ for all $g, h \in G$ and all n.
5. Prove that $(gh)^{-1} = h^{-1}g^{-1}$.
6. Prove that the intersection of two subgroups H and K of a group G is a subgroup of G.
7. Show that in the Abelian group G, the set $H = \left\{ g \in G \middle| g^n = e, \forall n > 0 \right\}$ is a subgroup.
8. Find all 5 of the distinct subgroups of Z_{16}.
9. Find all 8 of the distinct subgroups of the group Z_{24}
10. List all elements of $Z_{30}^{\times}$ and prove that they form a group.
11. Check that the collection of matrices g of the form $g = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ is a subgroup
12. Show that a group of order 101 has no subgroups except $\{e\}$ and itself.
13. Show that a group of order 1009 has no subgroups except for $\{e\}$ and itself.
14. For group G, where $g, h \in G$, show that if $\left|\langle g \rangle\right| = 30$ the and $\left|\langle h \rangle\right| = 77$, then $\left|\langle g \rangle\right| \cap \left|\langle h \rangle\right| = \{e\}$.
15. Prove that a group element and its inverse have the same order.
16. Check that $Z_8^{\times}$ cannot be generated by a single element.

Suggested Reading

- Garrett – Chapter 17
- Smart – Sections 1.1.1, 1.1.3, 1.1.2, A.6
- Forouzan – Section 4.1, pages 102-103, 254-258